# AIGC Technology and the Digital “Rebirth” of China's Fine Traditional Culture

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## Abstract

This article examines how artificial intelligence-generated content (AIGC) technology enables the digital “rebirth” of China's fine traditional culture. It first outlines the technical logic and model architectures of AI content production to identify its technological characteristics. It then analyzes AIGC's specific applications in producing and communicating traditional cultural content across four dimensions: textual generation and interpretation, multidimensional audiovisual presentation, the perception and construction of cultural spaces, and digitally interactive embodied experiences. Drawing on encoding/decoding theory, it explains the full process of digitally recreating China's fine traditional culture. Finally, it reflects on cultural misinterpretation, aesthetic rigidity, and commercial dilution in AIGC applications and proposes responses. As a new quality productive force for the digital transformation of China's fine traditional culture, AIGC should promote creative transformation while safeguarding culture's core, helping traditional culture combine cultural depth with responsiveness to the digital age.

Keywords: AIGC; China's fine traditional culture; cross-modal production; cultural heritage; reproduction of traditional culture

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## Introduction

General Secretary Xi Jinping has stated: “Without the flourishing and prosperity of Chinese culture, there can be no great rejuvenation of the Chinese nation. A nation's rejuvenation requires strong material forces as well as strong spiritual forces.”[1] China's fine traditional culture encompasses a broad range of art forms, rich artistic meanings, and a constellation of outstanding works, embodying the cultural accumulation of the country and nation over a long history. As informatization advances and scientific and technological capabilities grow, new technologies and creative methods are giving this culture renewed vitality.

With the launch of ChatGPT, the large AI model developed by OpenAI, in November 2022, the internet age entered its “first year of generative AI.” AI image-generation tools such as Midjourney and Stable Diffusion subsequently became widespread, followed by the emergence of numerous large video models. Multimodal generative AI (GenAI), spanning literature, painting, music, film, and television, has opened unlimited creative possibilities in the humanities while giving China's fine traditional culture new life in the digital age.

New media emerge through the convergence of two trajectories: the development of emerging media technologies and that of computing. As a contemporary intelligent media technology, AI can likewise be understood in relation to the development of other new media in the internet age. In *The Language of New Media*, Lev Manovich identifies numerical representation, modularity, automation, variability, and transcoding as principles of new media. AI's model construction and algorithmic learning exhibit numerical representation and modularity; content generation embodies automated production and variability across modalities. At the cultural level, AIGC's capacity for encoding and transcoding creates broader opportunities for expression across social environments.[2]

Within the treasury of China's fine traditional culture, how does AIGC integrate data and learn cultural essences through large models and algorithms? How can traditional cultural images be digitally represented in the AI age? How can AI carry forward and extend traditional culture in the AIGC era? The authors address these questions through AIGC examples involving poetry, painting, and other outstanding traditional artworks, alongside forms of intangible cultural heritage.

## I. Giving Life through Intelligent Computation: Deep Learning from Cultural Resources and Model Integration

### 1. The Nature and Technical Architecture of AIGC

AIGC (artificial intelligence-generated content) is a technology based on AI that enables computer systems to generate text, images, audio, video, and other forms of content automatically through model training and algorithmic optimization. Its core lies in using vast multimodal datasets to construct and optimize large deep neural networks, extracting data features through self-supervised and supervised learning to represent cultural meanings and enable innovation.

Within deep-learning architectures, mainstream models such as Transformers are commonly trained on large datasets to capture features of language, grammar, image texture, and color. From a semiotic perspective, the data processed through deep learning, a fundamental AI capability, can be understood as “signs carrying meaning.” Deep learning analyzes and reorganizes collections of signs, combining them into new semiotic texts.

In constructing large models, data preprocessing includes image denoising, edge enhancement, and the extraction of annotation information. Textual features are quantified through word embeddings and syntactic tree parsing. Adaptive neural network architectures, such as multilayer Transformers, are built according to the data and task. Their advantages include parallel processing and suitability for large-scale training. After training, models are further tuned using test sets, their performance evaluated, and feedback collected to optimize parameters continually. Through this systematic model iteration, AI gradually learns and reproduces the languages of Chinese culture and art.

Applying AI to the digital transmission of Chinese culture requires a departure from general-purpose training toward domain-specific model designs differentiated by cultural characteristics. This creates diverse frameworks for presenting traditional culture digitally. Taking account of distinct artistic categories, models should learn from and incorporate the rich expressive forms and artistic languages of traditional culture, balancing creative conventions and innovation. In literature, text-generation and conversational models can be customized for poetry, fiction, prose, and other genres. Poetry models, for example, emphasize rhythm and parallelism; fiction models focus on plot, characterization, and environmental description. Visual and audio models likewise address traditional painting, calligraphy, opera, and other audiovisual arts. Painting models generate digital images with distinctive artistic qualities in styles such as gongbi (meticulous painting) and xieyi (freehand expressive painting). Calligraphy

models imitate scripts ranging from archaic seal script to fluid running script, seeking to reproduce Chinese calligraphy's appeal digitally.

### 2. Domain-Specific Models and Diverse Frameworks

Unlike general-purpose models, large models for digitizing China's fine traditional culture require specialized, customized training frameworks tailored to cultural characteristics. Models supporting diverse artistic expressions can balance creative conventions with innovation. For example, the world's first Song-style hanfu model, released in 2023, combines image data and historical documents for multidimensional training, comprehensively learning and representing Song-style hanfu from color and texture to stylistic details. Training requires selective collection of relevant images, textual descriptions, historical records, artworks from related periods, and cultural classics. The data should include different garment styles, materials, colors, and wearing contexts so that the model can learn the full range of Song-style hanfu's characteristics.

At the platform level, AIGC's modular architecture provides flexible support for content generation. Alibaba Cloud's PAI ArtLab, for example, incorporates multiple models and lets users import data for training, enabling highly customized content-generation services. Its built-in data-flow management supports batch annotation, streamlining training and markedly improving efficiency. In March 2024, Zhejiang University's Art and Archaeology Image Data Laboratory used this platform to combine the characteristics of traditional Chinese painting with AIGC in its Ten Views of West Lake project, an industry-university-research collaboration described as China's first exploration of this combination.

AIGC platforms support creative production and communication in two ways. First, they manage and integrate large models and data resources. PAI ArtLab includes more than 100 mainstream models. Users can search for popular models and add them to “My Models” with one click, retrieving them as needed to generate content. They can also install plug-ins for additional AI functions. Second, platforms provide data-flow services, allowing users to upload workflows and save creative progress in real time. Uploaded dataset images can be automatically annotated in batches, preparing data for users' own task-specific models and improving the efficiency and accuracy of data processing.

## II. Regenerating Artistic Worlds: Automated Content Production and Cross-Modal Construction

Discussing automation in *The Language of New Media*, Lev Manovich observes that, from the nineteenth century onward, modern society developed technologies capable of automatically generating media. These accumulated unprecedented quantities of materials, including photographic collections, film archives, and audio archives. Media thereby entered a new stage in which technologies for storing, identifying, and rapidly retrieving media resources became necessary.[3] For Manovich, automation is an algorithmic process through which new media perform certain operations independently. As large models improve, generative AI relies on deep-learning frameworks and multimodal algorithms to generate different forms of content, moving beyond single modalities toward cross-modal production. AI is gradually transforming content production in the humanities and revitalizing traditional culture's expression in the contemporary era.

### 1. Textual Generation and Interpretation

At the textual level, AIGC's functions in presenting traditional culture can broadly be divided into two categories: assistance, with AI functioning as a tool; and creative subjectivity, with AI functioning as an artistic creator alongside humans.

As an assistive tool, AIGC encompasses the many ways general-purpose models help users understand traditional cultural knowledge and create artworks. These include disseminating cultural knowledge through human-machine textual dialogue and preparing prompts before AIGC creation.

Through natural language processing, AIGC models can turn complex traditional cultural concepts, historical events, and art forms into accessible, vivid explanations in response to instructions. They can also communicate cultural meanings through entertaining stories and interactive quizzes. Asked about the Tang poem *Quiet Night Thoughts*, for example, the Doubao model helps users understand it through a question-and-answer format.

The model asks: "What does '床' mean in '床前明月光'? A. A bed for sleeping; B. A well curb; C. A window."

It answers: "In ancient usage, '床' had several meanings. Here it is more likely to mean a well curb. Ancient well curbs were commonly made from stone or wood and resembled beds in shape. People often spent time near wells and could see the moon when they looked up, which is why Li Bai wrote '床前明月光'."

Within AIGC workflows, prompt design directly affects the precision of generated content. By incorporating specific context, styles, and constraints, prompts enable models to produce content matching creative requirements. This generation relies on the complex embedding layers of pretrained language models, such as the GPT series, which use self-attention to understand and generate text at a high semantic level. Users unfamiliar with AIGC or prompt-writing techniques can follow model suggestions or ask a model to write prompts, subsequently using these to direct that model or another model's output.

Poetry generation is also an important direction in natural language generation research within the digital humanities. In 2023, digital humanities teams at Nanjing Agricultural University and Nanjing Normal University developed and released the SikuGPT pretrained model, based on GPT-2, specifically for the automatic generation of classical Chinese texts. Their work addresses textual translation and comprehension, and they also released an open-source classical-to-modern Chinese translation model fine-tuned on bilingual parallel corpora. Developers of such classical poetry models chiefly evaluate three aspects of their output: grammatical fluency, textual coherence, and poetic meaning.[4] This article accordingly identifies two dimensions of deep learning in AIGC poetry creation: formal beauty and inner meaning.

Formally, AIGC poetry models generally begin by imitating classical Chinese conventions of meter, level and oblique tones, and rhyme. Using ERNIE 4.0, the authors supplied the opening line "孤山寺北贾亭西" and requested an original seven-character regulated verse (qiyan lüshi). The model generated the following:

孤山寺北贾亭西，（平平仄仄仄平平）

古木参天映碧溪。（仄仄平平仄仄平）

远岫含烟藏古刹，（仄仄平平平仄仄）

近湖泛棹荡涟漪。（平平仄仄仄平平）

钟声悠扬传幽谷，（平平平平平平仄）

*The source describes this line as a variant pattern that nevertheless satisfies the overall requirements of regulated verse.*

香火缭绕绕云霓。（仄仄平平仄仄平）

此间胜景何处觅，（仄平平仄平平仄）

*The source notes that the third character's level/oblique tone may be treated flexibly here.*

孤山寺里觅禅机。（平平仄仄仄平平）

In seven-character regulated verse, tonal patterns generally follow two-character rhythmic units, with level and oblique tones alternating. Regulated verse also requires carefully balanced parallelism in the second couplet (lines three and four) and third couplet (lines five and six). The authors therefore argue that AIGC clearly follows traditional metrical conventions in generating classical poetry, allowing its works to inherit classical Chinese poetry's rhythm and musical beauty.

At the level of content, creators using AIGC for poetry should give greater consideration to themes and imagery, inner meanings, and emotional expression to ensure artistic quality and beauty. In the SikuGPT2-poem model mentioned above, developers first collected classical poems on homesickness and reflections on the past, such as *Homesickness in Yangzhou* by the Song-dynasty poet Dou Yu:

家依楚水岸，身寄洋州馆。

望月独相思，尘襟泪痕满。

Inputting the poem's first line, "家依楚水岸," into the model produced:

家依楚水岸，门对杜陵坡。

日暮寒江口，蒹葭日晚汀。

Comparing the original and the AIGC output shows that the model mainly describes scenery and accumulates images. Dou Yu's original, by contrast, is filled with inward reflection and emotional expressions of longing. Pretraining should therefore emphasize humanistic expression and the creation of artistic meaning.

### 2. Multidimensional Presentation of Audiovisual Art

AIGC is widely used for open-ended painting and design, generating traditional Chinese styles such as ink painting and calligraphy from prompts. In costume design and film and television settings, it can precisely generate contemporary works incorporating traditional elements. Large models' digital transformation of traditional cultural content in audiovisual art focuses on several areas.

First, one of AIGC's most widespread applications is prompt-based painting and design organized around artistic styles and techniques. Users need only describe their proposed subject, style, and constituent elements. Models then learn material characteristics such as brushwork, color, technique, material, and texture to generate works. Baidu's Wenxin Yige AI art and creative assistance platform, built on its ERNIE models, for example, can understand and generate traditional Chinese styles such as gongbi and ink

painting, imitating their use of color and brushwork. The authors experimented with the prompt “Qi Baishi's shrimp, lively, Chinese ink-painting style” (Figure 1). The resulting work closely imitated and reproduced the source style's colors and forms. One shortcoming is that, because text and image generation have different mechanisms, Chinese character samples are unevenly distributed, and semantic structures are complex, most AIGC image models cannot accurately generate Chinese writing within images. They produce only visually similar symbols. AIGC has therefore yet to explore calligraphy in depth.



Figure 1. “Qi Baishi’s Shrimp,” generated by the Wenxin Yige model.

Beyond traditional painting, AIGC has been developed for digitally designing traditional cultural elements. The Song-style hanfu model mentioned earlier, for instance, designs historical clothing and contemporary garments incorporating ancient Chinese elements. Similarly, Alibaba's digital media and entertainment business released “Shenli Nishang,” its first specialized AI model for costume and styling design in film and television, chiefly serving period productions. Such models combine traditional culture and technology to improve the efficiency and professionalism of preproduction.

Second, AIGC empowers film and television, giving Chinese culture renewed vitality. In February 2024, China's first text-to-video animated series, Qianqiu Shisong (Odes to Poetry through the Ages), premiered on CCTV-1. Intended for children, it centers on poems in Chinese-language textbooks. Its first six episodes tell the stories of five Tang-dynasty poets, using AIGC to transform poetry into Chinese-style ink animation and present its sentiments toward family and nation. Its concise, lively storytelling suits contemporary rhythms and combines tradition with modernity, revitalizing ancient mythological stories in a new era.

AIGC can also function like the legendary Ma Liang's magic brush, restoring and creating digital twins of classic artworks and cultural artifacts. In May 2024, a 1.589-meter-high bronze statue of a kneeling human figure bearing a zun vessel, supported by an animal, was displayed at the Sanxingdui Museum in Guanghan, Sichuan. Excavated in separate archaeological areas at Sanxingdui, its components were joined

“across pits” with AI assistance. AI first calculated and analyzed the features of three-dimensional models to determine matching joins, then corrected deformations, filled missing areas to suggest the original appearance, and assisted in analyzing deterioration. An accompanying popular-science video demonstrated the restoration process, while line-drawing tools are planned to support future archaeological research. Beyond restoration, AIGC can digitally reproduce artifacts' original appearances through digital twins. Virtual artifacts can appear within audiovisual content, enabling audiences to inspect every detail of a vivid three-dimensional reconstruction and gain deeper knowledge of their cultural background.

### 3. The Perception and Construction of Cultural Spaces

AIGC introduces further possibilities for cultural exhibition spaces, chiefly through constructing immersive museum environments. It offers more vivid, multidimensional, and innovative forms of cultural communication and experience. In *The Production of Space* (1974), Henri Lefebvre connects physical, mental, and social space in a spatial triadic dialectic.[5] Traditional culture within museums begins in material space, the physical museum. As technology develops, virtual spaces supported by digital technologies gradually emerge, shifting the emphasis from physical display toward cultural exploration at the mental level.

In physical museums, AI uses immersive virtual environments to recreate and enhance traditional cultural displays. Augmented reality (AR), virtual reality (VR), and digital twins enable interaction between virtual and physical environments, deepening visitors' understanding of historical scenes and cultural atmospheres. Ahead of China's 2024 “Two Sessions,” for example, the Chengjiang Fossil Site Natural History Museum used AIGC text-to-video technology to let visitors “return” to the Cambrian world and encounter ancient organisms across time. This immersive experience presented the period's distinctive ecology and life forms, demonstrating an innovative combination of history and modernity, science and culture.

In online cultural spaces, immersive virtual environments built on AIGC bring the digitization of traditional culture into the metaverse era. The Palace Museum's digital projects, including Panorama of the Palace Museum, Famous Paintings of the Palace Museum, and the Digital Treasure Gallery, offer vivid interpretations of its culture through AI in virtual space. The panorama project, for instance, uses AI to optimize snow-scene rendering, image enhancement, and intelligent guide recommendations. AIGC allows cultural presentation to transcend temporal and spatial limits, digitally and intelligently preserving and communicating Palace Museum culture to audiences worldwide and advancing its transmission and development.

### 4. Digital Interaction and Embodied Experience

The generativity and emergence of large AI models bring an interactive transformation to digital content. Consumers of cultural products and viewers of artworks gain greater freedom and choice, including the possibility of exchanging ideas and conversing with works. Their role gradually shifts from passive recipients of information toward producers and leading participants in its emergence. Within traditional Chinese culture, AIGC transforms one-way communication into embodied user experience, particularly through interactive design in digital content with traditional cultural themes.

Virtual characters, or digital humans, are a frontier application of AIGC in cultural communication. Combining computer graphics, deep learning, and natural language generation (NLG), they use image synthesis, real-time rendering, and speech synthesis to acquire natural expressions and vocal abilities, interacting with users in real time. In response to AIGC's development, China's National Radio and Television Administration has issued policies including the recommended broadcasting, television, and online audiovisual industry standard Technical Requirements for Digital Virtual Humans.[A] These

measures encourage the integration of digital humans and traditional arts, creating opportunities for their use in cultural transmission and communication.

A deeper application appears in character interaction in Chinese digital games, including generated plots and scenes. NPC interaction shifts from conventional linear narrative exchanges to emergently generated content, providing players with freer and richer gameplay. The knowledge resources in large datasets also enable multilayered cultural understanding. Justice Mobile (Nishuihan) is a Chinese pioneer in combining AIGC and gaming. Its NPCs are trained chiefly on wuxia fiction, historical classics, and poetry, songs, and literary rhapsodies. They can converse with players and be actively approached and won over, forming deeper relationships. Players explore an immersive historical-cultural setting while establishing intellectual resonance and emotional connections with digital humans.

## III. Cultural Derivation: Encoding, Decoding, and Recreating China's Fine Traditional Culture

In “Encoding and Decoding in the Television Discourse,” Stuart Hall proposes that cultural production and communication can be understood as encoding and decoding. At the level of meaning, broadcasting structures must construct encoded messages in the form of meaningful discourse.[6] The institutional-social relations of production can become products only through linguistic forms. Before a message produces an “effect,” satisfies a “need,” or is put to “use,” it must first be apprehended as meaningful discourse and meaningfully decoded. These decoded meanings produce effects, exert influence, entertain, instruct, or persuade, leading to conceptual, cognitive, ideological, or behavioral consequences. At one moment, a structure uses codes to produce a message; at another, that message enters a structure through different modes of decoding. Encoding and decoding are therefore necessary steps in the digital regeneration of traditional culture.

In the generative AI era, culture is digitally encoded into meaningful codes and incorporated into digital cultural information systems. It is then communicated through advanced digital media and decoded, shaping new forms of communication and reception and strengthening audiences' identification with, and interest in, traditional culture. On this basis, and through the reshaping of determining forces such as social values, creative practices, and audiences' psychological needs in contemporary cultural contexts, traditional culture is both transmitted and extended, rewritten and recreated.

Digital encoding takes different forms according to the nature of the cultural work. First, for tangible visual arts such as historical artifacts and sculpture, it chiefly depends on high-resolution image capture, laser scanning, and three-dimensional reconstruction to preserve geometry, color, and texture. Through data annotation and model training, combined with three-dimensional convolutional networks and generative models, AIGC enables faithful virtual reconstruction. The Dunhuang Academy's virtual museum, for example, uses AIGC to capture and encode mural colors, lines, and patterns and sculptural shapes and textures, translating them into specific data structures within digital models. Dunhuang's cultural heritage can thereby be preserved comprehensively and in detail, providing rich data for subsequent communication and research.

Second, encoding abstract traditional cultural content, such as stories, myths, and classical texts, takes the form of AI-generated, perceptible digital artworks and images. CCTV's AI-produced microdrama series AI Looks at the Classics encodes traditional Chinese myths and stories from the extensive classical corpus. Through close understanding and encoding, it transforms dry textual narratives into vivid microdramas and visualizes their cultural ideas.

Third, intangible cultural heritage transmitted as artistic skills and creative methods is encoded through AIGC analysis of large bodies of work. This preserves existing cultural materials while learning artistic languages and techniques and constructing models that bring heritage into machine-assisted creation. In the digital performance Feimeng (Not a Dream), presented at the Suzhou Museum's "Echoes: AI Dialect Art Exhibition," techniques including motion capture, high-precision recording, AI processing, and digital scanning recreate and generate people, voices, and scenes. The result is a dream space combining virtual and real elements for narrative reconstruction and Kunqu opera duets.

Cultural decoding is a further rebirth after encoding. Technology carries cultural life forward while giving audiences more dimensions of knowledge, perception, and emotional resonance. In *Understanding Media: The Extensions of Man*, Marshall McLuhan argues that media shape and control the scale and form of human association and action.[7] As an emerging medium, AIGC is reshaping traditional culture's communication and reception. Cultural transcoding can take many forms, including cultural landscapes combined with other advanced digital media that communicate traditional culture's meanings and spirit. A VR version of the Yuju opera The Seventh-Rank Magistrate, released in Zhengzhou in September 2023, remade the classic through VR. AI-assisted three-dimensional spatial synthesis faithfully reconstructed the historic Xun County magistrate's office as it appeared 600 years ago, allowing viewers to explore it interactively. Decoding traditional cultural elements through virtual reality enables younger generations to experience them immersively and gives traditional culture renewed vitality.

Transmitting traditional culture does not mean rigid imitation. It requires continual integration with contemporary values and spiritual concerns, and sustained development, so that traditional culture can be inherited and extended in the new era, opening new chapters in Chinese culture.

## IV. Challenges and Reflections

Although AIGC has achieved results in enabling traditional culture's digital rebirth, it faces challenges and dilemmas requiring reflection.

First, large AIGC models may misinterpret traditional culture. Limitations in data sources and corpus quality can produce bias and partial readings, causing outputs to depart from genuine cultural meanings. Such misinterpretation appears first at the level of outward artistic form and technique: a prompt for an "ink-painting portrait," for instance, may produce a three-dimensional-style figure. It also occurs at the content level, where models lack sensitivity to artistic meaning and grasp only superficial elements. If the following lines are supplied to an AIGC video model:

> 春风又绿江南岸，明月何时照我还

it may depict only landscape images such as "spring breeze," "green trees," and "riverbank," without using visual narrative to evoke the poet's state of mind and emotional atmosphere.

Second, AIGC's aesthetic styles show limitations and rigidity when generating traditional cultural content. This is closely connected with training mechanisms and data distributions. Traditional Chinese aesthetics is a diverse, dynamically developing system encompassing rich styles, expressive techniques, and aesthetic ideas. Yet most models' deep-learning samples concentrate on a limited selection of classic or popular styles, producing convergence. Many AI ink paintings, for example, follow similar compositions and techniques, lacking precise understanding and innovative interpretation of the distinctive styles of different historical periods and regional schools. Over time, such homogenization causes aesthetic fatigue and prevents traditional culture's deeper essence and subtle artistic meanings from being fully presented.

Moreover, capital-driven commercialization of cultural products is pronounced in the AIGC era. Catering to vulgar market tastes can erode the seriousness and coherence of traditional cultural themes. In some religious and other cultural domains, for example, AIGC lacks clear norms and systematic model training, and outputs may conflict with the culture's established conventions. Relevant policies and regulations should therefore be introduced, platform content review strictly implemented, and these problems addressed from the top down. Although technology creates new means and opportunities for cultural development, it also introduces risks and challenges. Standardized governance of AIGC should be strengthened, alongside improvements in public cultural literacy and aesthetic judgment, to foster an environment conducive to traditional culture's healthy development in the digital age.

## Conclusion

This study examines pathways for integrating AIGC and China's fine traditional culture. By analyzing technical architectures, application scenarios, and communication mechanisms, it reveals AIGC's central value as a new quality productive force enabling traditional culture's digital transformation. Further research is nevertheless needed. First, domain-specific adaptation in certain artistic categories remains insufficiently explored, leaving the technological transformation of less prominent cultural forms inadequately covered. Second, cultural misinterpretation, aesthetic rigidity, and commercial dilution require urgent attention. Productive interaction between AI and culture can be advanced through domain-specific models, cultural content review mechanisms, and collaboration among industry, universities, and research institutions. With policy guidance, scholarly inquiry, and public participation working together, China's fine traditional culture can preserve its spiritual roots while acquiring contemporary vitality, becoming cultural wealth that transcends time and space.

## Notes and References

Poetry quotations, model-generated verses, and Chinese tonal patterns are retained in Chinese as requested. “平” denotes a level tone and “仄” an oblique tone. Tonal scans and model responses are reproduced as presented in the article; explanatory comments are translated. English titles of cultural projects and Chinese-language articles are descriptive translations unless an established English title is used. The original's reference numbering is retained.

{A} National Radio and Television Administration. Technical Requirements for Digital Virtual Humans: GY/T 411-2024. November 28, 2024. Accessed November 10, 2025. https://www.nrta.gov.cn/art/2024/11/28/art_113_69707.html

[1] Chinese Government Website. Xi Jinping, “Speech at the Forum on Literature and Art.” October 15, 2024. Accessed April 5, 2025. https://www.gov.cn/yaowen/liebiao/202410/content_6980503.htm

[2, 3] Lev Manovich. *The Language of New Media*. Translated into Chinese by Che Lin. Guiyang: Guizhou People's Publishing House, 2020, pp. 79 and 97, respectively.

[4] Liu Jiangfeng, Liu Chufei, Qi Yue, Liu Liu, Li Bin, Liu Chang, and Wang Dongbo. “Exploring AIGC-Assisted Digital Humanities Research in Practice: Classical Chinese Poetry Generation Driven by SikuGPT.” Information Studies: Theory & Application, 2023, no. 5, pp. 23-31. In Chinese.

[5] Henri Lefebvre. *The Production of Space*. Translated into Chinese by Liu Huaiyu. Beijing: The Commercial Press, 2021, p. 10.

[6] Stuart Hall. “Encoding and Decoding in the Television Discourse.” Translated into Chinese by Xiao Shuang. Shanghai Culture, 2018, no. 2, pp. 33-45, 106, 125-126.

[7] Marshall McLuhan. *Understanding Media: The Extensions of Man*. Translated into Chinese by He Daokuan. Nanjing: Yilin Press, 2019, p. 40.

## Terminology and Translation Notes

This section records translation choices and checks; it is separate from the authors' argument.

“China's fine traditional culture” consistently renders the article's central phrase. “New quality productive forces” follows the established policy term. “Multimodal” refers to multiple data or expressive modalities; “cross-modal” emphasizes movement and transformation between them. “Gongbi,” “xieyi,” “hanfu,” “Kunqu,” “Yuju,” and “wuxia” retain culturally specific art and genre names, with contextual explanations where needed.

Manovich's five principles are rendered as numerical representation, modularity, automation, variability, and transcoding, following his English terminology. Hall's encoding/decoding framework is distinguished from Manovich's cultural transcoding. Standard author and work-title forms are Lev Manovich, *The Language of New Media*; Henri Lefebvre, *The Production of Space*; Stuart Hall, “Encoding and Decoding in the Television Discourse”; and Marshall McLuhan, *Understanding Media: The Extensions of Man*. Quotations translated from the cited Chinese editions have not all been matched word for word against the original English or French editions.

Terminology and titles were checked against Manovich's own “New Media from Borges to HTML” (https://manovich.net/content/04-projects/033-new-media-from-borges-to-html/30_article_2001.pdf), MIT Press's *The Language of New Media* (https://mitpress.mit.edu/9780262632553/the-language-of-new-media/) and Understanding Media (https://mitpress.mit.edu/9780262631594/understanding-media/), Duke University Press's Writings on Media (https://www.dukeupress.edu/writings-on-media), and Wiley's *The Production of Space* (https://www.wiley-vch.de/en/areas-interest/humanities-social-sciences/the-production-of-space-978-0-631-18177-4). The virtual-human standard was checked against the issuing authority's notice cited in note A.